Le novità di SQL Server 2019

Gianluca Hotz

Who am I?

- Gianluca Hotz | @glhotz | ghotz@ugiss.org
- Independent Consultant
  - 20+ years on SQL Server (from 4.21 in 1996)
  - Database modeling and development, sizing and administration, upgrade and migration, performance tuning
- Interests
  - Relational model, DBMS architecture, Security, High Availability and Disaster Recovery
- Community
  - 20 years Microsoft MVP SQL Server (from 1998)
  - Founder and President UGISS
    - User Group Italiano SQL Server (PASS Chapter)

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Configuration

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Installation

- Most new things in 2016/2017
  - Separate downloads
  - IFI and tempdb configuration
  - Linux
  - R and Python integration
  - Integration Services Scale-out
  - Polybase
- New best practices added in 2019
  - MaxDOP configuration
  - Min/Max Server memory configuration

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SQL Server 2019 new DBE configuration options

SQL Server 2019 RC1 Setup

Database Engine Configuration

Specify Database Engine authentication security mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and Filestream settings.

- Product Key
- License Terms
- Global Rules
- Microsoft Update
- Product Updates
- Install Setup Files
- Install Rules
- Feature Selection
- Feature Rules
- Instance Configuration
- PolyBase Configuration
- Java Install Location
- Server Configuration
- **Database Engine Configuration**
- Integration Services Scale Out ...
- Integration Services Scale Out ...
- Consent to install Microsoft R ...
- Consent to install Python
- Feature Configuration Rules

Server Configuration | Data Directories | TempDB | MaxDOP | Memory | FILESTREAM

When an instance of SQL Server runs on a computer that has more than one CPU logical core, it detects the best degree of parallelism, that is, the number of processors employed to run a single statement, for each parallel plan execution. MAXDOP specifies the maximum number of cores to utilize for this instance.

Detected logical CPU cores on this computer: 4

Maximum degree of parallelism (MaxDOP)*: 4

** The displayed default value was either calculated by Setup, or was explicitly specified on the Setup command line with the /SQLMAXDOP parameter.*

You can modify the MaxDOP here to be used as the default in all query executions for this instance, unless overridden at the query level. To suppress parallel query plan generation, set MaxDOP to 1.

See Configure the max degree of parallelism Server Configuration Option for more information.

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SQL Server 2019 RC1 Setup

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Server Configuration | Data Directories | TempDB | MaxDOP | Memory | FILESTREAM

SQL Server can change its memory requirements dynamically based on available system resources. However, in some scenarios you can configure the range of memory (in MB) that is managed by the SQL Server Memory Manager for this instance, by specifying min server memory and/or max server memory.

	Recommended	Default
Min Server Memory (MB):	0	0
Max Server Memory (MB):	512	2147483647

** The displayed recommended values were calculated by Setup based on your system configuration and edition, unless these were explicitly specified in the Setup command line using the /SQLMINMEMORY and /SQLMAXMEMORY parameters.*

For more information see: Server Memory Server Configuration Options.

☐ Click here to accept the recommended memory configurations for the SQL Server Database Engine

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PolyBase Java Connector for HDFS (< CTP 3.2)

SQL Server 2019 CTP2.2 Setup

Feature Selection

Select the Developer features to install.

- Product Key
- License Terms
- Global Rules
- Microsoft Update
- Product Updates
- Install Setup Files
- Install Rules
- **Feature Selection**
- Feature Rules
- Instance Configuration
- PolyBase Configuration
- Server Configuration
- Database Engine Configuration
- Consent to install Microsoft R ...
- Consent to install Python
- Feature Configuration Rules
- Ready to Install
- Installation Progress
- Complete

Looking for Reporting Services? Download it from the web

Features:

- Instance Features
  - [x] Database Engine Services
    - [x] SQL Server Replication
    - [x] Machine Learning Services (In-Database)
      - [x] R
      - [x] Python
    - [ ] Full-Text and Semantic Extractions for Search
    - [ ] Data Quality Services
    - [x] PolyBase Query Service for External Data
      - [x] Java connector for HDFS data sources
  - [ ] Analysis Services
- Shared Features
  - [ ] Machine Learning Server (Standalone)
    - [ ] R
    - [ ] Python
  - [ ] Data Quality Client

Feature description:

Includes PolyBase Java Connector that enables truly integrated querying across HDFS data using standard T-SQL statements.

Prerequisites for selected features:

- Microsoft MPI v9

Needs to be manually installed:

- Microsoft Visual C++ 2012 Redistributable
- Oracle SE Java Runtime Environment Version 7 Update 51 or higher (64-bit)

Disk Space Requirements

Drive C: 11509 MB required, 116830 MB available

Select All | Unselect All

Instance root directory: C:\Program Files\Microsoft SQL Server\

Shared feature directory: C:\Program Files\Microsoft SQL Server\

Shared feature directory (x86): C:\Program Files (x86)\Microsoft SQL Server\

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PolyBase Java Connector for HDFS (CTP 3.2+)

SQL Server 2019 RC1 Setup

Feature Selection

Select the Developer features to install.

- Product Key
- License Terms
- Global Rules
- Microsoft Update
- Product Updates
- Install Setup Files
- Install Rules
- **Feature Selection**
- Feature Rules
- Instance Configuration
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- Server Configuration
- Database Engine Configuration
- Consent to install Microsoft R ...
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- Ready to Install
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Looking for Reporting Services? Download it from the web

Features:

- Instance Features
  - [x] Database Engine Services
    - [x] SQL Server Replication
    - [x] Machine Learning Services and Language Extensions
      - [x] R
      - [x] Python
      - [x] Java
    - [ ] Full-Text and Semantic Extractions for Search
    - [ ] Data Quality Services
    - [x] PolyBase Query Service for External Data
      - [x] Java connector for HDFS data sources
  - [ ] Analysis Services
- Shared Features
  - [ ] Machine Learning Server (Standalone)
    - [ ] R
    - [ ] Python
  - [ ] Data Quality Client
  - [ ] Client Tools Connectivity
  - [ ] Integration Services

Feature description:

Includes a set of technologies for copying and distributing data and database objects from one database to another and synchronizing between the databases for consistency. You can use replication to distribute data to different locations, and to remote and mobile users over local and wide area networks, dial-up connections, wireless connections and the Internet.

Prerequisites for selected features:

Already installed:
- Windows PowerShell 3.0 or higher

To be installed from media:
- Microsoft Visual C++ 2017 Redistributable
- Microsoft MPI v10

Disk Space Requirements

Drive C: 11661 MB required, 116088 MB available

Select All Unselect All

Instance root directory: C:\Program Files\Microsoft SQL Server\

Shared feature directory: C:\Program Files\Microsoft SQL Server\

Shared feature directory (x86): C:\Program Files (x86)\Microsoft SQL Server\

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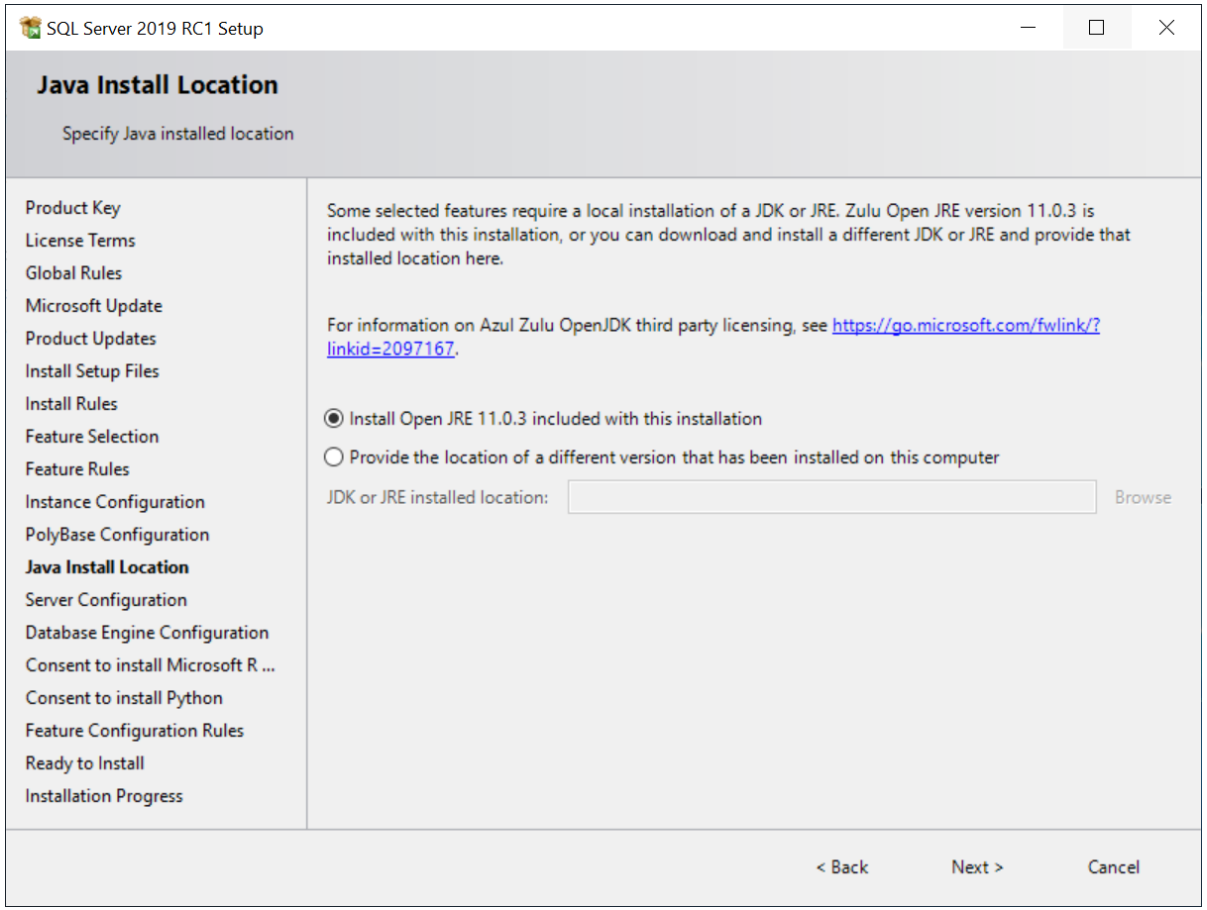
Free Supported Java

- Azul System's Zulu OpenJDK
  - Starting from CTP 3.2+
  - No need to install separately
  - Polybase, Spark, Java Extensibility
- Free support
  - Jointly by Microsoft & Azul
  - Open Source
- Other Java flavors supported



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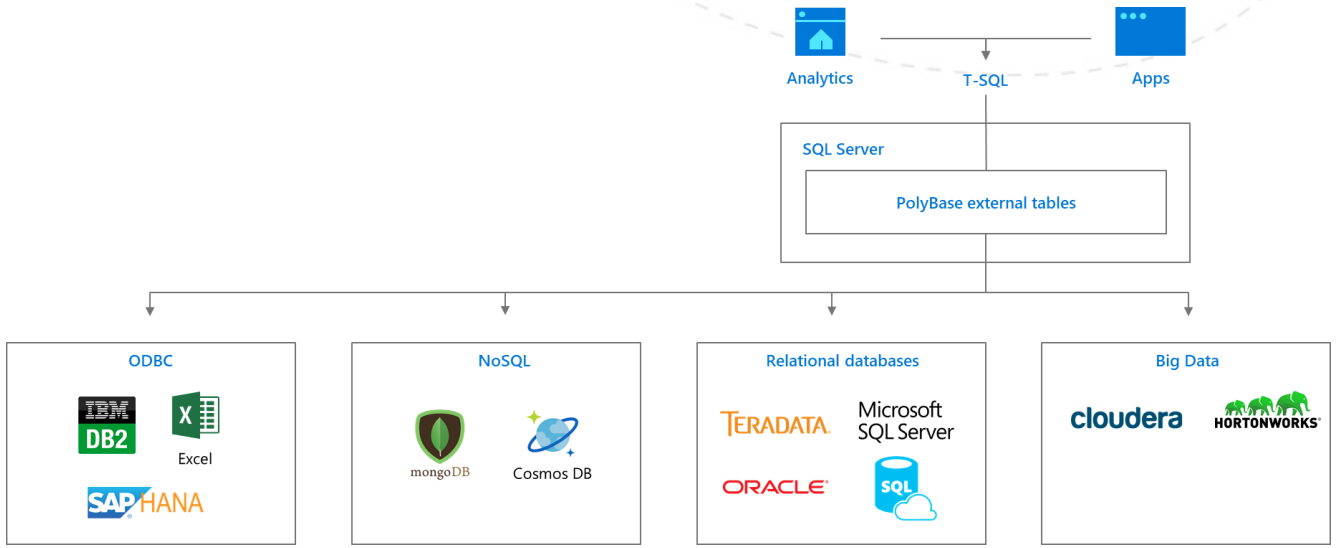
Polybase

- Data Virtualization
  - External Data Source
  - External Tables
- SQL Server 2016+
  - Azure Blob Storage, Hadoop
- SQL Server 2019
  - SQL Server (all flavors), Oracle, Teradata, MongoDB, CosmosDB (Mongo API)
  - ODBC Generic Type (e.g. DB2, SAP Hana, Excel, ...)



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External Tables vs Linked Servers

External Tables

- Database Scoped
- ODBC Drivers
- Read-only*
- Scale out queries with push-down
- Failover with AG
- Basic authentication
- Distributed Transactions not supported

Linked Servers

- Instance Scoped
- OLEDB Providers
- Read/write
- Single threaded queries with push-down
- Requires separate config from AG
- Basic and integrated authentication
- Distributed Transactions supported

* Insert into HDFS allowed

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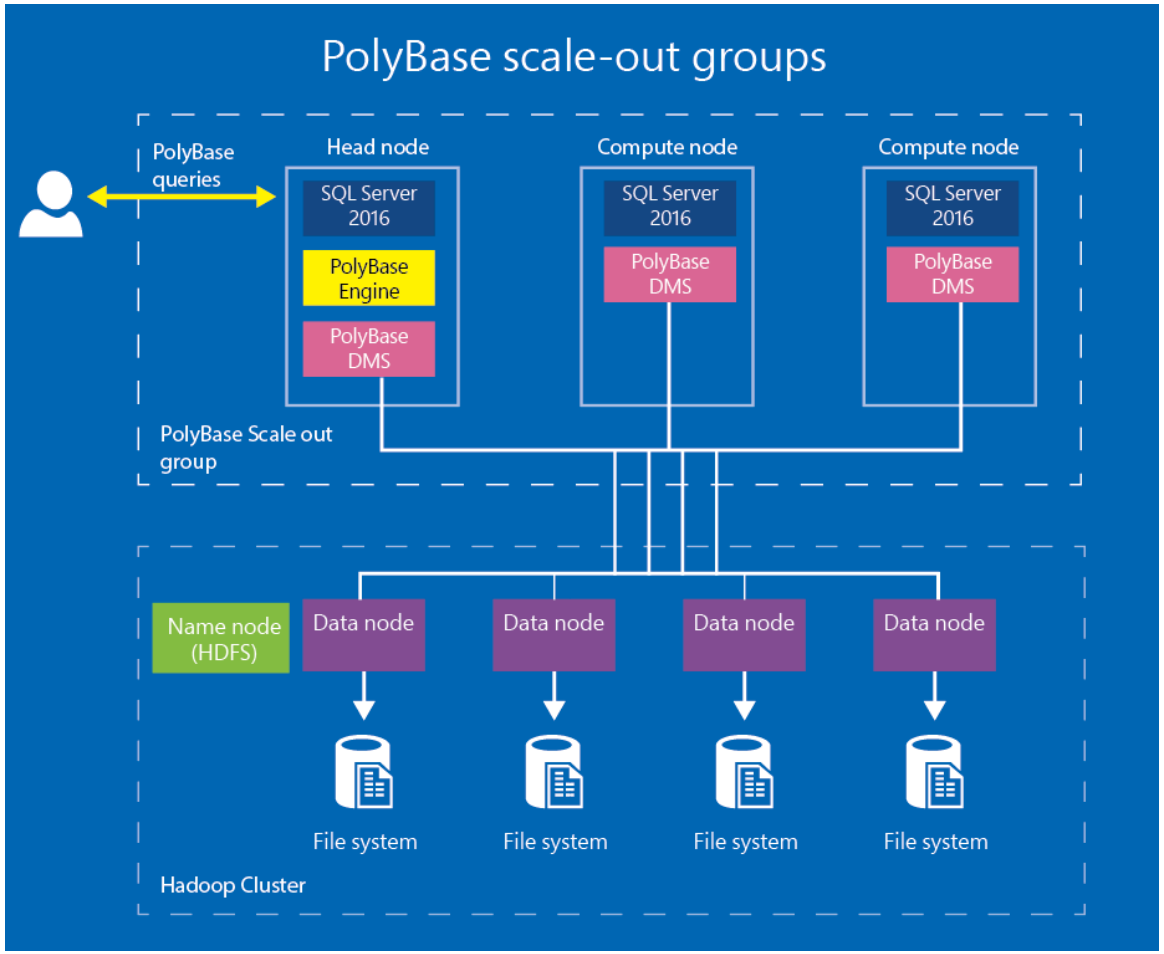
Polybase Scale-Out Groups



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SQL Server on Linux

- Replication
  - Snapshot, Transactional & Merge
  - No immediate update subscribers, no Oracle publishing
- Microsoft Distributed Transaction Coordinator (MSDTC)
- OpenLDAP support for third-party AD providers
- Machine Learning Services
- Change Data Capture
- Polybase

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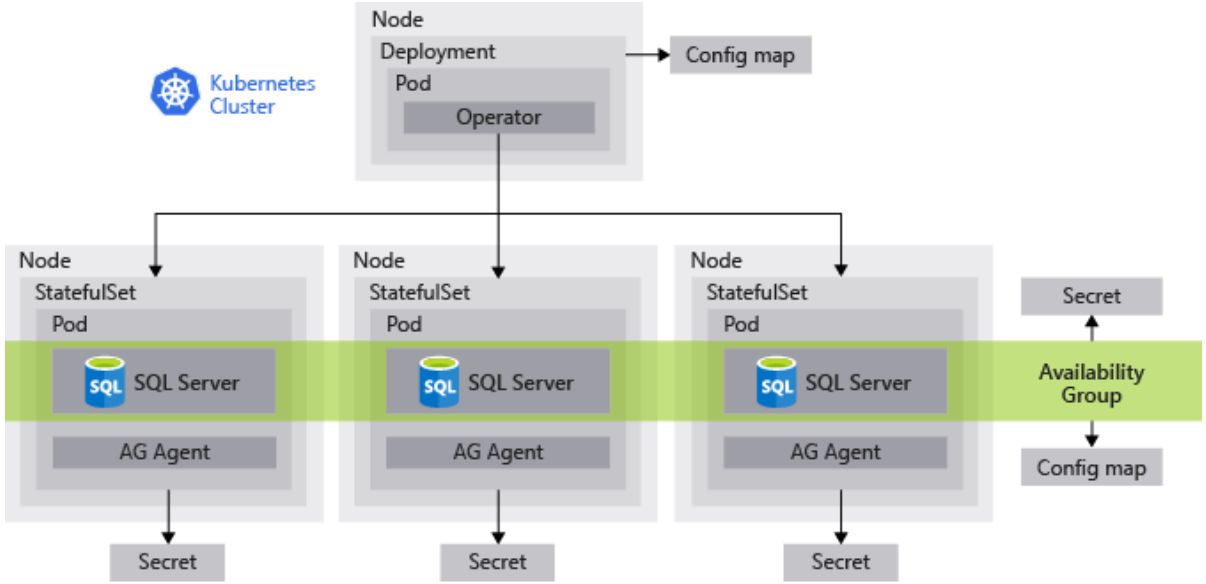
Containers

- New container registry
- Always On Availability Group on Docker containers with Kubernetes
  - **<u>Not supported as of RC1 and not available in RTM</u>**



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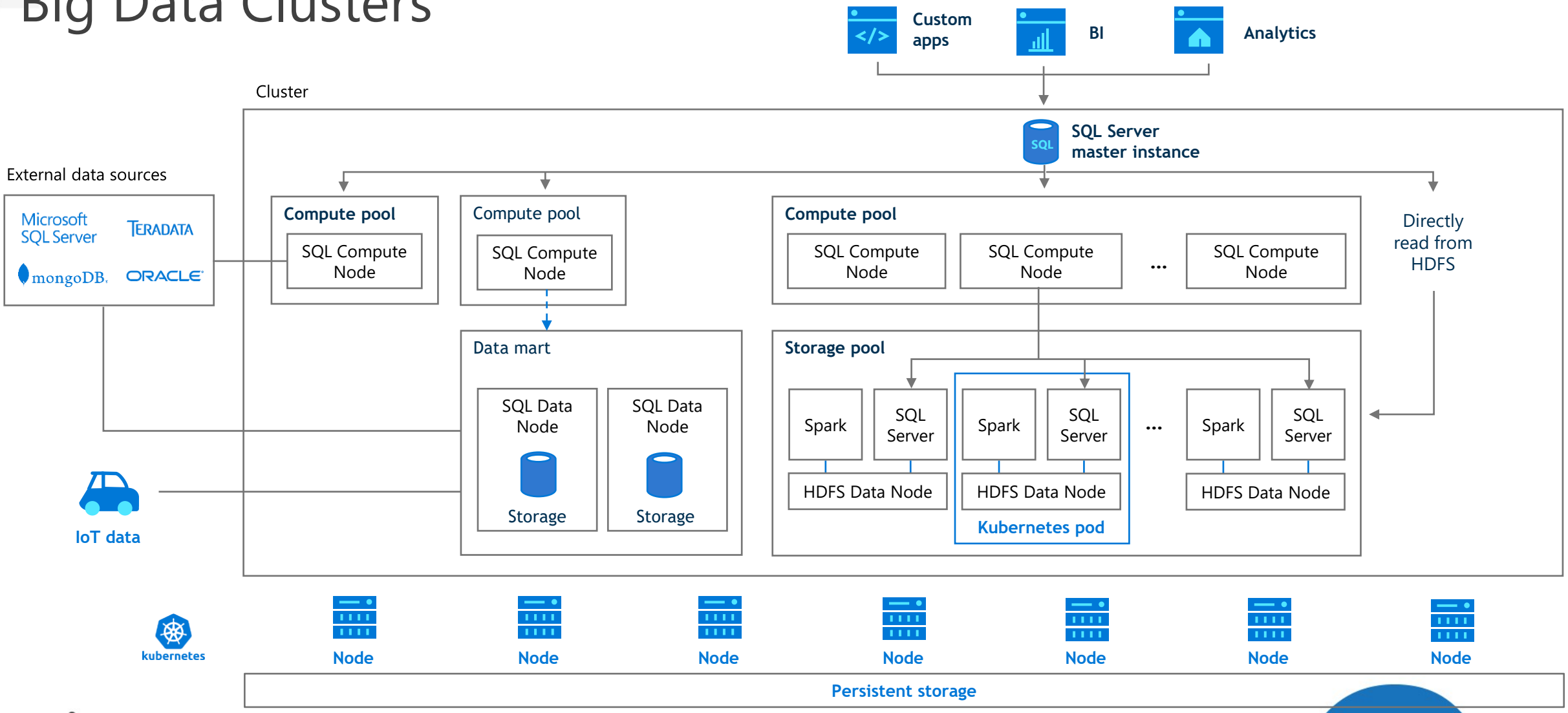
Big Data Clusters



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Administration

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Tools

- SQL Server Management Studio V18
- Azure Data Studio (was Operations Studio)
  - Windows, MacOS, Linux
  - Azure SQL Database, Azure SQL Data Warehouse, PostgreSQL
  - SQL Server 2019 extensions including Big Data Cluster (Hadoop etc.)
  - Notebooks

Resumable online index operations

- Resume after failure (e.g. out of disk space)
- Pause and resume later (e.g. free temporarily resources)
- Manage large indexes using less log an shorter transactions
- Fit rebuild operations into limited maintenance windows
  - ```
    REBUILD
    WITH (ONLINE = ON, RESUMABLE = ON, MAX_DURATION = 30
    MINUTES);
    ```
- SQL Server 2017+ Resumable Index Rebuild
- SQL Server 2019+ Resumable Index Create

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    ```

Transparent Data Encryption

- ALTER DATABASE … SET ENCRYPTION **SUSPEND**
- ALTER DATABASE … SET ENCRYPTION **RESUME**

Columnstore Indexes

- **Online** build/rebuild Clustered Columnstore
  - E.g. conversion from row-store to Columnstore

DBCC CLONEDATABASE

- Instantaneous schema-only copy of a database for troubleshooting
  - No data, full-schema, statistics and Query Store
  - Non-blocking
  - Read-only by default (can be changed)
  - Optionally **NO_STATISTICS**, **NO_QUERYSTORE**
  - SQL Server 2012 SP4, 2014 SP2 CU3, 2016 SP1, 2017
- New in SQL Server 2019
  - Columnstore Statistics

Accelerated Database Recovery

- Benefits
  - Fast and consistent database recovery
    - Number/size of active transactions don't impact recovery time
  - Instantaneous transaction rollback
    - Active time and number of updates don't impact rollback time
  - Aggressive Log Truncation
    - Even with long running transactions, prevents growing out of control
- High level
  - Versioning all physical database modifications
  - Only logical operations undone (are limited and can be undone instantly)
  - Active transactions at crash time are marked as aborted
  - Versions generated by aborted transactions can be ignored by user queries
- Currently available in Preview in Azure SQL Database and CTP2.3+!

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Improved diagnostic data for stats blocking

- Query Waiting for synchronous update operations
- Now **sys.dm_exec_requests** shows **SELECT (STATMAN)**
- New **WAIT_ON_SYNC_STATISTICS_REFRESH** wait stat

session_id	request_id	start_time	status	command
59	0	2018-11-13 11:50:36.617	suspended	SELECT (STATMAN)

wait_type	waiting_tasks_count	wait_time_ms	max_wait_time_ms	signal_wait_time_ms
WAIT_ON_SYNC_STATISTICS_REFRESH	1	18781	18781	0

Availability Groups Enhancements

- More synchronous replicas
  - SQL Server 2012 4 replicas, 2 synchronous
  - SQL Server 2014 8 replicas, 2 synchronous
  - SQL Server 2017 8 replicas, 3 synchronous
  - SQL Server 2019 8 replicas, **5 synchronous**
- Secondary-to-primary read/write redirection
  - READ_WRITE_ROUTING_URL and ApplicationIntent=ReadWrite (default)
  - Killer feature to replace Listener
    - Cluster technology not offering listener-like features
    - Multi-subnet scenarios too complex to setup/maintain (e.g. Pacemaker)
    - Read scale-out or DR with cluster type NONE

Storage Class Memory / PMEM

- Allows low latency I/O
  - memory-mapped memcpy-like operations in user mode
- SQL Server 2016 SP1
  - NVDIMM-N for tail of the log caching
- SQL Server 2019
  - PMEM devices Linux
  - support for data, log In-Memory OLTP checkpoint files placement
- Hybrid Buffer Pool
  - Clean pages direct referenced on PMEM devices without copy
  - Dirty pages still kept in DRAM

Estimating Data Compression savings

- **sp_estimate_data_compression_savings**
  - Returns specified object's current size and estimates
- SQL Server < 2019
  - ROW and PAGE compression
- SQL Server 2019+
  - Adds **COLUMNSTORE** and **COLUMNSTORE_ARCHIVE** compression
  - Object type determines Columnstore type
    - E.g. Heap -> Clustered, Clustered index -> Clustered

Query Profiling Infrastructure

- Run-time information on query execution plans
  - E.g. live query statistics
- Lightweight profiling
  - introduced in SQL Server 2014 SP2, 2016 SP1+
  - 2% expected CPU overhead vs. 75%
  - SQL Server 2019+
    - Enabled by default
    - XEvents **query_post_execution_plan_profile** (vs **query_post_execution_showplan**)
    - DMF **sys.dm_exec_query_plan_stats()** last known ***actual*** execution plan

Internal database pages information

- Undocumented DBCC PAGE
- New in SQL Server 2019
  - **sys.dm_db_page_info(DatabaseId, FileId, PageId, Mode)**
  - **page_resource** column in **sys.dm_exec_requests** and **sys.sysprocesses**
  - **sys.fn_PageResCracker(page_resource)** to get **db_id, file_id, page_id**

Results | Messages

	database_id	file_id	page_id	page_header_version	page_type	page_type_desc	page_type_flag_bits	page_type_flag_bits_desc	page_flag_bits	page_flag_bits_desc
1	2	1	1	1	11	PFS_PAGE	0x0		0x0	
2	2	1	2	1	8	GAM_PAGE	0x0		0x0	
3	2	1	3	1	9	SGAM_PAGE	0x0		0x0	
4	2	1	4	NULL	NULL	NULL	NULL	NULL	NULL	NULL
5	2	1	5	NULL	NULL	NULL	NULL	NULL	NULL	NULL
6	2	1	6	1	16	DIFF_MAP_PAGE	0x0		0x200	HAS_CHECKSUM
7	2	1	7	1	17	ML_MAP_PAGE	0x0		0x0	
8	2	1	8	1	1	DATA_PAGE	0x4		0x200	HAS_CHECKSUM
9	2	1	9	1	13	BOOT_PAGE	0x0		0x0	
10	2	1	10	1	10	IAM_PAGE	0x0		0x200	HAS_CHECKSUM
11	2	1	11	1	2	INDEX_PAGE	0x4		0x200	HAS_CHECKSUM

Query Store Enhancements

- Capture mode AUTO is now the default
- Support for custom capture mode e.g.
  - QUERY_CAPTURE_MODE = **CUSTOM**,
    QUERY_CAPTURE_POLICY = (
    STALE_CAPTURE_POLICY_THRESHOLD = 24 HOURS,
    EXECUTION_COUNT = 30,
    TOTAL_COMPILE_CPU_TIME_MS = 1000,
    TOTAL_EXECUTION_CPU_TIME_MS = 100
)

Development

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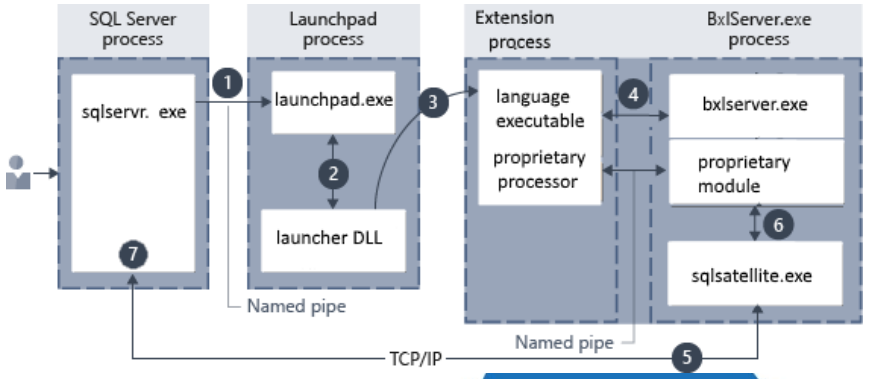
Truncation error messages

- Error message 8152 too generic
  - String or binary data would be truncated
- SQL Server 2019 introduces message 2628
  - String or binary data would be truncated in **table** '%.*ls', **column** '%.*ls'. Truncated **value**: '%.*ls'
- Enabled with Trace Flag 460
  - Opt-in required to avoid breaking parsing applications

Extensibility Framework

- Secure external script execution
- Scale/optimization opportunities
- SQL Server integration (e.g. store procedures, PREDICT)
- Language Support
  - SQL Server 2016+ Support for R
  - SQL Server 2017+ Support for Python
  - SQL Server 2019+ Support for Java



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UTF-8 Support

- Full support for import/export, collations, replication, …
  - Still not for Linked Servers, In-Memory OLTP, External Table (Polybase)
- CHAR and VARCHAR support (Windows collations only)
- UTF8 in collation names
  - E.g. LATIN1_GENERAL_100_CI_AS_SC_UTF8
- Can provide storage savings
  - E.g. up to 50% from NCHAR(10) to CHAR(10) with UTF8 vs UTF16

SQL Graph enhancements

- Derived tables and view support in MATCH queries
  - Set of nodes/edges using UNION ALL
  - Useful for heterogeneous entities or connections between them
- MATCH support in MERGE
- Edge Constraints
  - CONNECTION constraint
- SHORTEST_PATH function in MATCH
  - Shortest path between two nodes/entities
  - Single source shortest path
  - Shortest path from multiple source to multiple destinations

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Security

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Certificate Management

- Extended in SQL Server Configuration manager
  - View and validate certificates installed
  - View certificates close to expiration
  - Deploy Certificates across machine in Availability Groups
  - Deploy Certificates across machine in Failover Cluster Instances

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SQL Server Sensitivity Classification

- SQL Server 2019+
- (already available in Azure SQL Database)
- T-SQL command **ADD|DROP SENSITIVITY CLASSIFICATION**
  - applies to tables, columns
  - **LABEL**, **LAtBEL_ID**, **INFORMATION_TYPE**, **INFORMATION_TYPE_ID**
- Metadata stored in **sys.sensitivity_classifications**
- SQL Server Audit add column **data_sensitivity_information**

SQL Data Discovery and Classification

- Discovery & Recommendations, Labeling, Reporting
- SQL Server management Studio Tool (V17.5+)
- Support for SQL Server 2008+ and Azure SQL Database
- Metadata con be persisted and queried
  - Based on Extended Properties in older versions
    - sys_information_type_name, sys_sensitivity_label_name
  - Based on SQL Server Sensitivity Classification in newer versions

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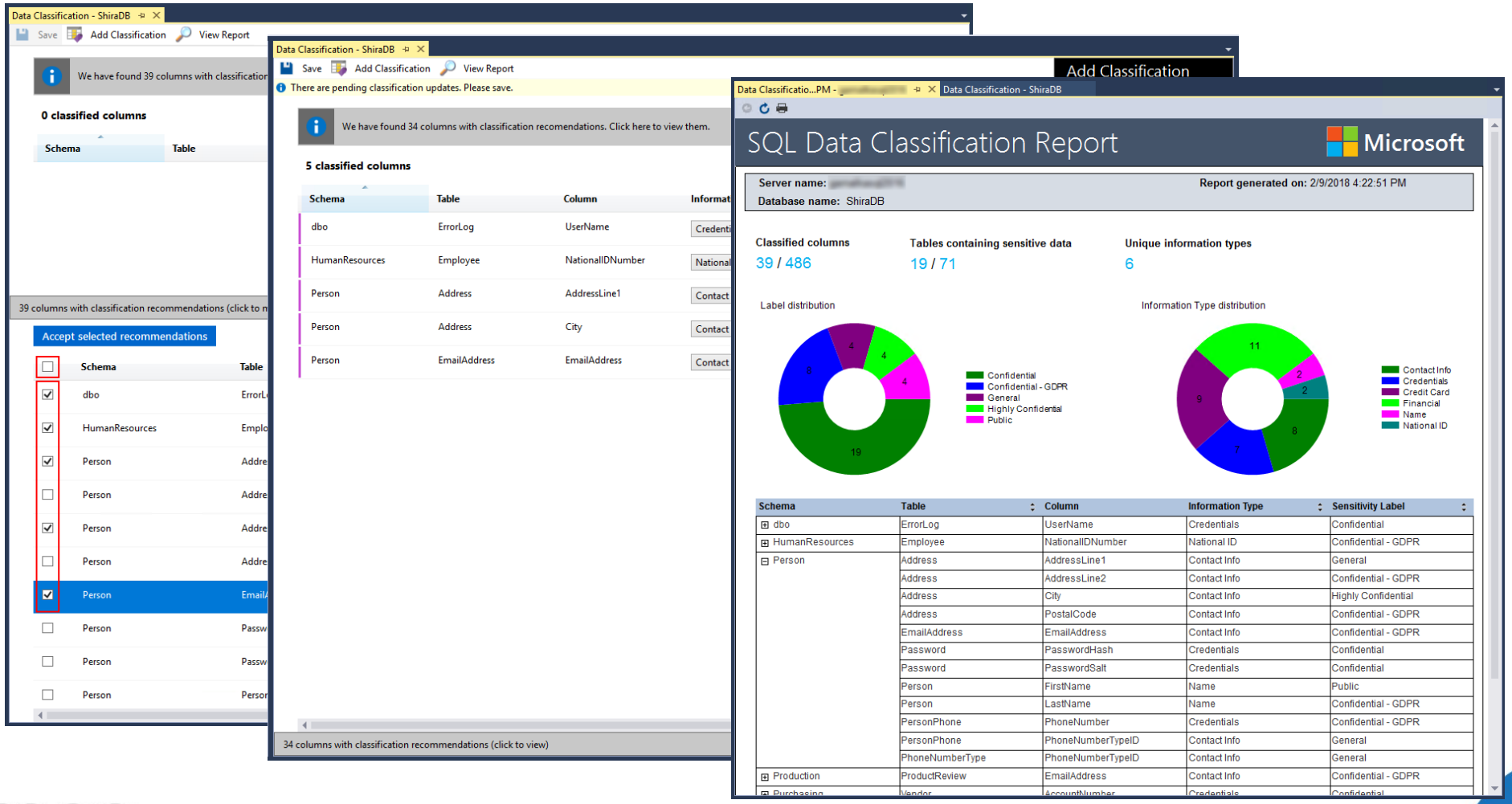
Data Discovery and Classification Demo



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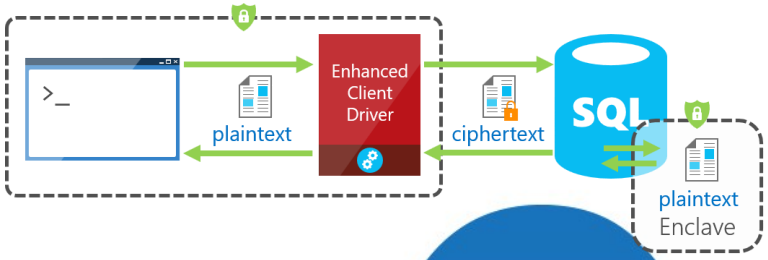
Always Encrypted with Secure Enclaves

- Basic architecture as SQL Server 2016+ implementation
- Now allows server-side computation on encrypted columns
  - In-Place Encryption (ALTER TABLE for initial encryption)
  - Rich computations (e.g. range comparisons, LIKE predicates, indexing, …)
- Inside secure enclaves
  - Virtualization-based Security (VBS) secure memory enclaves
    - also known as Virtual Secure Mode(VSM) enclaves
  - Operation on plaintexts cannot be disclosed outside enclave
  - Column Master Keys sent over secure channel by client driver
- Still some limitations
- Performance optimizations pending…



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Performance

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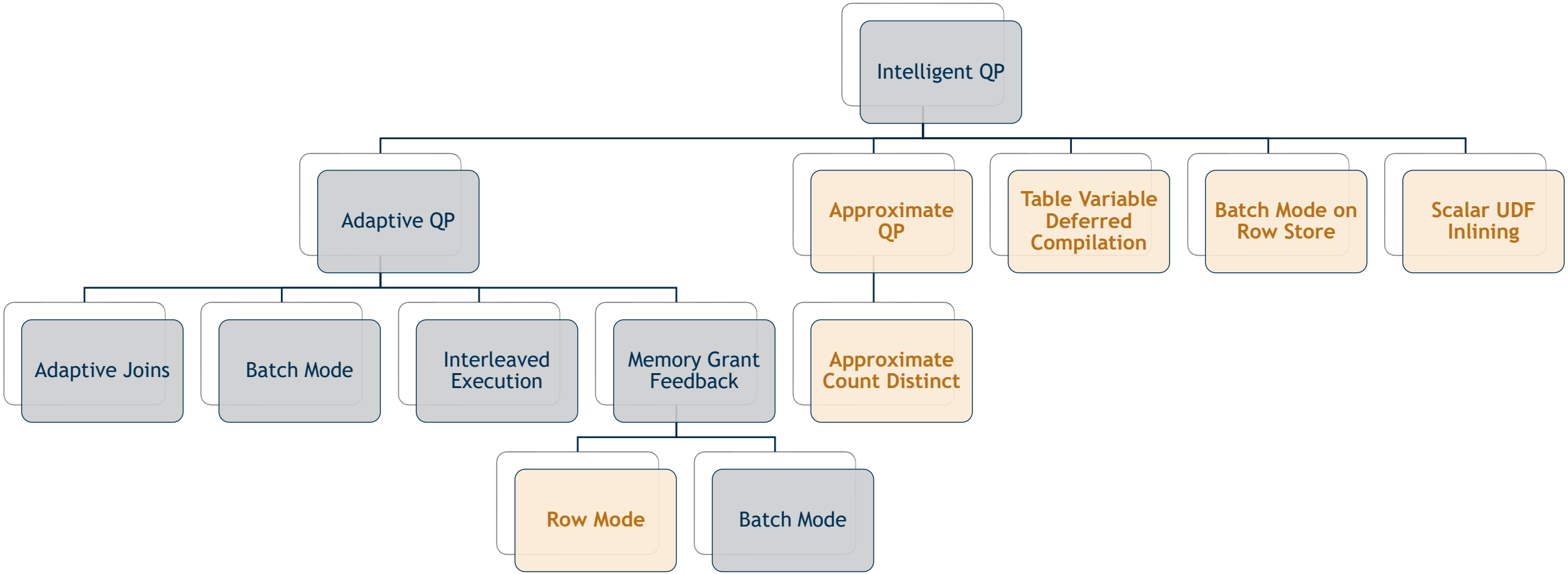
Sequential Keys Optimization

- Common problem inserting
  - PAGELATCH_EX waits on last page in the b^+-tree
  - https://support.microsoft.com/en-us/help/4460004
- New index option
  - OPTIMIZE_FOR_SEQUENTIAL_KEY
  - Addresses “convoy” specifically problems
  - Latency may be higher but stable throughput
    - New wait type BTREE_INSERT_FLOW_CONTROL

Memory-Optimized Tempdb Metadata

- ALTER SERVER CONFIGURATION
  SET MEMORY_OPTIMIZED TEMPDB_METADATA = ON
- Limitations
  - Instance restart needed
  - Tempdb system tables not accessible by transactions using other memory-optimized tables in user databases
  - Locking hints on tempdb system tables not honored
- Issues with Columnstore indexes on temporary tables (fixed by RTM?)

Intelligent Query Processing



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Execution Modes

- Row Mode
  - Execution tree iterators consume 1 row at a time
  - Traditional execution mode for Rowstore
- Batch Mode
  - Execution tree iterators consume a batch of rows at a time
  - Optimal with large scan operations (e.g. large table aggregates or joins)
  - SQL Server 2012 introduced to leverage Columnstore Indexes
  - SQL Server 2016/2017 extended usage scenarios for CI
  - SQL Server 2019 extended usage scenarios to Rowstore

Batch Mode on RowStore

- Help reducing CPU Consumption
- Columnstore still a better choice
  - for OLAP workload that is I/O bound
  - can’t always create it (e.g. impact on OLTP, features not supported)
- Limitations
  - In-Memory tables not supported (only heaps & disk-based b-trees)
  - Not used when fetching/filtering LOB columns
  - (including sparse columns sets & XML)

Batch Mode on Rowstore Control

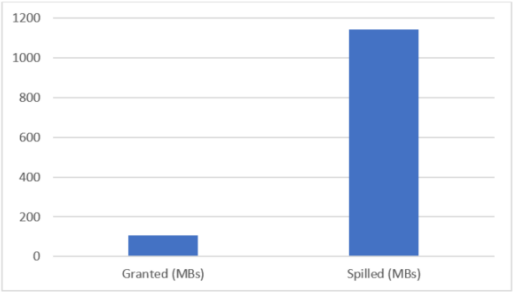
- SQL Server < 2019
  - Some scenarios covered with tricks… (article part1, part2, part3)
- SQL Server 2019+
  - Scenarios supported directly by Query Processor
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION SET **BATCH_MODE_ON_ROWSTORE** = ON|OFF
  - OPTION (USE HINT ('**ALLOW_BATCH_MODE**'));
  - OPTION (USE HINT ('**DISALLOW_BATCH_MODE**'));

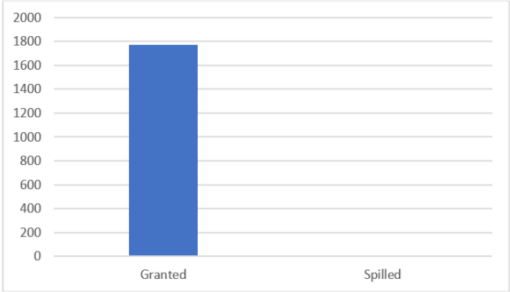
Memory Grant

- Excessive Grant
  - Too much memory allocated vs. memory used
  - Impact: blocking, out-of-memory, reduced concurrency
- Poor Grant
  - Not enough memory allocated resulting in data spill to tempdb
  - Impact: slow query, excessive disk usage (tempdb)
- Grant increase
  - dynamic grants increase allocation too much
  - impact: server instability, unpredictable performance

Memory Grant Feedback

- Post-execution evaluation
  - Updates grant value for cached plan
  - E.g. more memory if spilled, less if excessive grant
- Version support
  - SQL Server 2017+ Batch Mode
  - SQL Server 2019+ Row Mode
- Plan caching
  - Not persistent (i.e. not saved in Query Store)
  - OPTION(RECOMPILE) prevents caching and memory grant feedback





Memory Grant Feedback Control

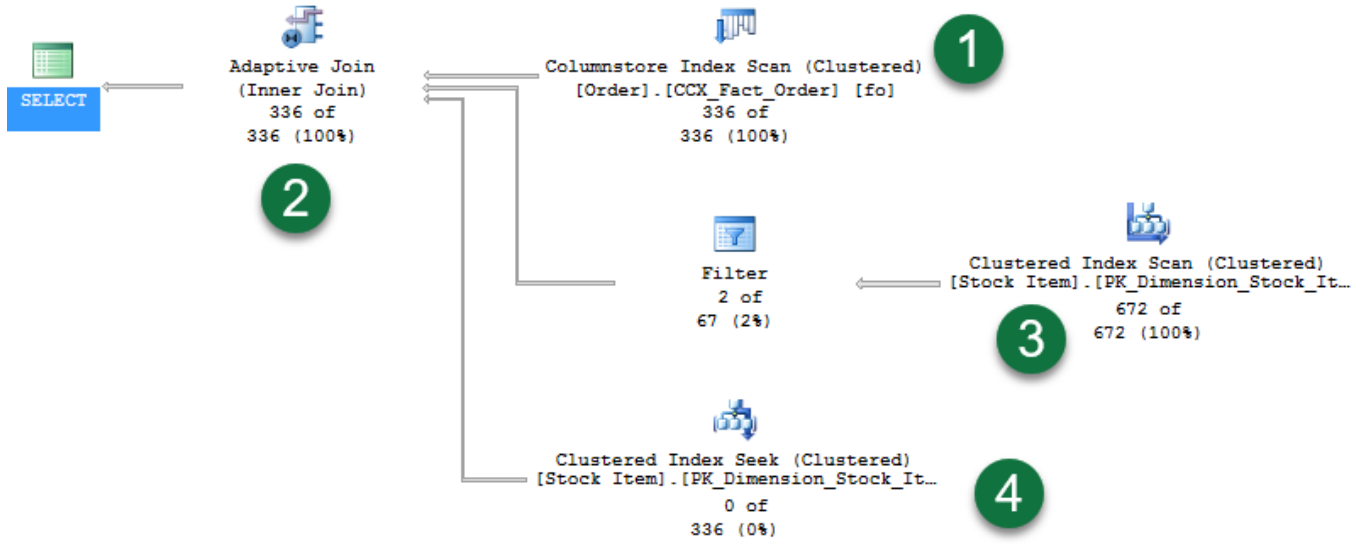
- Batch Mode
  - On by default with database compatibility level **140+**
  - ALTER DATABASE SCOPED CONFIGURATION SET **BATCH_MODE_MEMORY_GRANT_FEEDBACK** = ON|OFF
  - OPTION (USE HINT('**DISABLE_BATCH_MODE_MEMORY_GRANT_FEEDBACK**'));
- Row Mode
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION SET **ROW_MODE_MEMORY_GRANT_FEEDBACK** = ON|OFF
  - OPTION (USE HINT ('**DISABLE_ROW_MODE_MEMORY_GRANT_FEEDBACK**'));

Troubleshooting Memory Grant Feedback

- Parameter sensitive scenarios
  - Some queries requires different plans with different grants
  - Memory grant feedback will disable itself when unstable
- Extended Events to monitor changes
  - SQL Server 2017+ **memory_grant_feedback_loop_disabled**
  - SQL Server 2019+ **memory_grant_updated_by_feedback**
- SQL Server 2019+ execution plan attributes
  - **IsMemoryGrantFeedbackAdjusted**
    - No: First Execution, Accurate Grant, Feedback disabled
    - Yes: Adjusting, Stable
  - **LastRequestedMemory**

Batch mode adaptive joins

- Scenario
  - Nested loop algorithm better for small build join inputs
  - Hash algorithm better for bigger inputs
- Adaptive joins defer choice after first input scanned



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Interleaved Execution

- Problem with multi-statement table valued functions (MSTVFs)
  - SQL Server <= 2012 optimize with cardinality = 1
  - SQL Server 2014 & 2016 optimize with cardinality = 100
- SQL Server >= 2017
  - Start optimization
  - Pause and executes MSTVFs if candidate
  - Resume optimization with correct cardinality

Table Variable vs Temporary Tables

Area	Temporary Tables	Table Variables
Manual statistics creation and update	Yes	No\
Indexes	Yes	Only inline index definitions allowed
Constraints	Yes	Only PRIMARY KEY, UNIQUE and CHECK
Automatic statistics creation	Yes	No
Creating and using a temporary object in a single batch	Compilation of a statement that references a temporary table that doesn't exist is deferred until the first execution of the statement	A statement that references a table variable is compiled along with all other statements before any statement that populates the Table Variable is executed, so compilation sees it as 1

Table Variable Deferred Compilation

- Before SQL Server 2019
  - Statement referencing TV compiled before population
  - Number of row estimate fixed at 1
- Starting with SQL Server 2019
  - Behaves like Temporary Tables
  - Statement referencing non existing TV is deferred until first execution
  - Number of row estimate much better
- Control
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION
    SET **DEFERRED_COMPILATION_TV** = ON|OFF
  - OPTION (USE HINT ('**DISABLE_DEFERRED_COMPILATION_TV**'));

Scalar UDF inlining

- T-SQL user defined functions that returns a single data value
- Performance problems
  - Iterative invocation
    - once per row, context switching especially with query execution
  - Lack of costing
    - before, only relational operators were costed, assumption to be cheap…
  - Interpreted execution
    - each statement executes in isolation, no cross-statement optimizations
  - Serial execution
    - Intra-query parallelism not allowed

Scalar UDF Automatic inlining

- In SQL Server 2019 Scalar UDF automatically transformed into
  - Scalar Expressions
  - Scalar Subqueries
- Optimize the whole plan (UDFs no longer visible)
- Control
  - On by default with database compatibility level **150+**
  - ALTER DATABASE SCOPED CONFIGURATION
    SET **TSQL_SCALAR_UDF_INLINING** = ON|OFF
  - OPTION (USE HINT ('**DISABLE_TSQL_SCALAR_UDF_INLINING**'));
  - CREATE FUNCTION … WITH **INLINE** = ON | OFF

Scalar UDF inlining example

```
CREATE FUNCTION dbo.discount_price(@price DECIMAL(12,2), @discount DECIMAL(12,2))
RETURNS DECIMAL (12,2) AS BEGIN RETURN @price * (1 - @discount); END

SELECT L_SHIPDATE, O_SHIPPRIORITY
, SUM(dbo.discount_price(L_EXTENDEDPRICE, L_DISCOUNT))
FROM LINEITEM, ORDERS
WHERE O_ORDERKEY = L_ORDERKEY
GROUP BY L_SHIPDATE, O_SHIPPRIORITY
ORDER BY L_SHIPDATE
```

10GB CCI compressed TPC-H Schema, 2 x CPUs (12 cores), 96GB RAM, SSD storage

	Query without UDF	Query with UDF (no inlining)	Query with UDF (inlining)
Execution time	1.6 seconds	29 minutes 11 seconds	1.6 seconds

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Scalar UDF inlining requirements

- Written using the following constructs
  - DECLARE, SET (var declaration/assignments)
  - SELECT (single/multiple var assignments)
  - IF/ELSE (arbitrary nesting levels)
  - RETURN (single or multiple)
  - UDF nested/recursive function calls
  - Relational operations like EXISTS, ISNULL
- No invocation of functions that are
  - time-dependent (GETDATE())
  - has side effects (NEWSEQUENTIALID())
- Uses EXECUTE AS CALLER (default)
- No table variables references
- No table-valued parameters references
- No user-defined types references
- Not natively compiled
  - interop supported
- Not a partition function
- Not referenced in
  - GROUP BY clauses
  - computed columns
  - check constraints
- No signatures added to it

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Scalar UDF inlining troubleshoting

- Column **is_inlineable** in **sys.sql_modules**
  - Doesn’t imply it will always be inlined! (e.g. 1000s lines of code)
- Execution Plan
  - If inlined successfully, xml node **<UserDefinedFunction>** will be missing
- Extended Events
  - **tsql_scalar_udf_not_inlineable**

APPROX_COUNT_DISTINCT

- Returns **approximate** number of unique non-null values in groups
  - HyperLogLog algorithm guarantees
    up to 2% error rate within 97% probability
- Fast data exploration with low memory footprint
  - E.g. dashboards, trend analysis, feature selection, etc.
  - Think 10 billion rows,
    1 user using 1,5GB memory vs 100 users using 12MBs
  - Tradeoff: precision, only scenarios where exact values are not necessary!

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Demo

- Scalar UDF inlining
- APPROX_COUNT_DISTINCT

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- - - - Thank you - - -

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